

[WORKING TITLE: Paid Access: Test-Time Compute on a Conservative Memory as a Physically-Metered Resource]

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Abstract

Test-time compute — retries, escalation, non-local routing — is usually a black box bolted onto a trained model: more forward passes, a learned halting head, no account of what capability the extra compute buys or what it costs the model’s guarantees. We take a **position**: when the underlying memory is a *conservative* (symplectic) associative memory, every mechanism that buys capability at inference can be made to carry an explicit **physical receipt** — a phase-volume Jacobian, a bounded energy ledger, an exact latch-transport law — and the resulting statements are **theorems about the mechanism, not empirical scaling curves**. We develop this “paid access” frame for a single unit whose latent state (q, p) is advanced by a damped symplectic (velocity-Verlet) step of a learned Hamiltonian, and split reachability into two provably distinct failure modes: **reach** (the target lies outside a kinematic *causal box* C_T , half-widths $L_i = T\epsilon c/\sqrt{M_i}$, set by velocity not energy) and **escape** (inside the box, behind a barrier). A **Lorentz squeeze** cures escape with a bounded, governor-re-absorbable injection $\leq e^{2|\zeta|}H$ and *prices reach rather than capping it* (its instantaneous displacement $p_0 \sinh \zeta/M_0$ carries the state beyond the energy-blind box, but at an energy cost growing exponentially in rapidity); an **intra-unit wormhole** (a gated canonical translation) cures reach by teleporting across the box at $\det J = 1$ *exactly*, paying only a *fixed* discrete ledger $\Delta V = V(b) - V(a)$ independent of distance and **transporting** any latched content by an exact $p^\top X \Delta$ — the load-bearing dichotomy as a **pricing law**: squeeze reach is exponentially priced in distance, wormhole reach is flat-priced. On a designed analytic testbed (dim 2 and 4, 5 seeds, oracle placement, laptop-CPU) we *verify* the full certificate stack — squeeze reach steps up and is then priced out past the swept rapidity budget $\zeta \leq 2.0$ (its reach is the bracket $[L, L + p_0 \sinh \zeta/M_0]$, not a knife-edge; the same bracket predicts $\zeta \approx 2.01$ to reach $d = 4.0$), the wormhole lands flat at all distances with $\det J = 1$ and ledger = 0 exact, and a Newtonian-mode control reaches past the box (confirming the cap is the constraint). The receipt is not a label: on this designed testbed it separates **transport** from **state-replacement**. The wormhole’s canonical translation *transports* a stored Goldstone charge by the exact $p^\top X \Delta$ (spread preserved, $\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}}) = 0.0803$); an untrained **state-replacing** map $(q, p) \mapsto (b, p)$ — a no-physics baseline that reaches by overwriting the state — has $\det J = 0$ *exactly* (measured) and **erases** it ($\text{std}(Q_{\text{out}}) = 0$). Deciding *whether* to take a certified edge is orthogonal: a learned decision head (§4.2) routes *through* the $\det J = 1$ channel and inherits its receipt — the certificate prices only transport. The fine print travels with the claim: volume preservation *alone* is not the latch receipt (a $\det J = 1$ random shift also scrambles the charge — the *matched channel* is what preserves it), and a bounded — even *free* — energy ledger does not buy BIBO (a $\Delta H = 0$ exit into a non-coercive region escapes with $r^* \propto T$; coercive-*component* membership is the operative clause). We then situate three *honest* pillars on **learned** memories, each scoped and priced in-line: a distribution-free calibrated compute-rationing gate whose mechanism is **memory-agnostic**; a direct one-hop non-local edge whose cost is flat in the number of units where multi-hop diffusion scales; and a settled regime

map on which a modern Hopfield network is the cheaper *and* more cue-noise-robust retriever at matched accuracy, the CLU gate reaching Hopfield accuracy only for clean/correlated cues at small capacity. The contribution is the **certificate stack and its discipline**, not a benchmark win.

1 Introduction

Adaptive test-time compute is a standard lever: models spend more inference budget on harder inputs via learned halting (ACT, Graves 2016; PonderNet, Banino et al. 2021), confidence-gated early exit (CALM, Schuster et al. 2022), learned routing (Mixture-of-Depths, Raposo et al. 2024; Mixture-of-Experts, Shazeer et al. 2017), and energy-as-verifier “thinking” (Energy-Based Transformers, Gladstone et al. 2025). Across this literature the gate is a *learned scalar* bolted onto a feedforward stack, and “more compute” means *more of the same operation* with no structural guarantee on what that operation does to the model’s state.

This paper takes a different starting point and a **position**. When the memory being queried is a *conservative* dynamical system — a latent state (q, p) advanced by a **structure-preserving symplectic integrator** of a learned separable Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with optional per-step damping supplying controllable forgetting — then test-time compute is **access to a phase space**, and the physics of that phase space *prices every access mechanism explicitly*. We call this **paid access**: each mechanism carries a **physical receipt** — the Jacobian determinant $\det J$ (did it conserve phase volume?), a bounded energy ledger (how much did it inject, and where does that energy go?), and a latch-transport law (what did it do to content stored in a protected direction?). The receipts are **certificates that follow from the symplectic structure, not curves fit to runs**. Our thesis is that this is the right way to reason about test-time compute on a conservative memory, and it is falsifiable: we state each certificate as a theorem and verify it, to machine precision, on a designed testbed.

The reference unit. Our reference memory is the **CLU (Causal Learning Unit)**, introduced as **CHLU in Jawahar & Pierini (2026)**; the exactly-solvable theory whose certificates we verify is developed in a companion note (Anonymous, 2026; hereafter *the theory note*). Nothing below uses a property specific to the reference training objective except where stated. **Nomenclature (do not conflate):** the learned **inertial mass** M_i (diagonal of T ; larger $M \Rightarrow$ slower, smaller light-cone) and the **spectral mass** μ_k (a normal-mode stiffness) run in opposite directions. Reach statements use M ; retention statements use μ .

Contributions.

1. **The paid-access frame and its two failure modes (§2–3).** Reachability splits into *reach* (kinematic; target outside the causal box C_T , $L_i = T\epsilon c/\sqrt{M_i}$) and *escape* (energetic; inside, behind a barrier); the two are cured by different mechanisms with different receipts. [*proven; theory note + Anonymous 2026.*]
2. **The certificate stack, verified (§3, headline).** On a designed testbed (dim 2/4, 5 seeds, oracle placement, laptop-CPU) the squeeze cures escape with bounded injection $\leq e^{2|\zeta|}H$ and *prices reach* (its reach is the bracket $[L, L + p_0 \sinh \zeta/M_0]$ — it exceeds C_T only by its instantaneous displacement, at energy growing exponentially in ζ); the wormhole cures reach at $\det J = 1$, a *fixed* ledger = 0 exact, transporting the latch by exact $p^\top X \Delta$; a dense potential fails reach; a Newtonian control confirms the cap. [*verification of the theory’s exactness; oracle placement.*]

3. **The receipt has a measured downstream consequence (§3.2.1, headline panel).** The certificate is not a decorative label. An untrained **state-replacing** map $(q, p) \mapsto (b, p)$ has $\det J = 0$ *exactly* (measured by autodiff, not asserted) — volume-annihilating and non-invertible — so it collapses a capture ball of stored states onto a single charge, **erasing** the Goldstone spread ($\text{std}(Q_{\text{out}}) = 0$, all 16 probes bit-identical), where the wormhole **transports** it ($\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}}) = 0.0803$; $\Delta Q = p^\top X \Delta$ to 1.2×10^{-7} ; reconstruction exact to 2.2×10^{-8}). **Fine print travels with the claim:** volume alone is *not* the latch receipt (a $\det J = 1$ random shift also scrambles the charge), and a bounded — even free ($\Delta H = 0$) — ledger is *not* sufficient for BIBO (§3.1, App. B.2). [*verification; designed testbed, oracle placement.*]
4. **Calibrated compute-rationing on an escalatable learned memory (§4.1).** A self-calibrating gate rations relaxation budget on a *trained* memory ($4.81 \pm 0.44 \times$ fewer relaxation steps **at kv16, at a 400-epoch budget**; the payoff decays to $1.57 \times$ at kv24 and $1.14 \times$ at kv32; never below always-full at any level; 5 seeds, MQAR vocab-256, laptop), with Learn-then-Test coverage certificates (30/30 valid). The *saving* is the robust invariant; the *accuracy gain* over always-full is a 400-ep (not-yet-converged) property — at convergence (§4.3, App. C.4.a) the gate’s accuracy **matches** full-budget, and the payoff is rationing, not accuracy. The gate mechanism is **memory-agnostic**; the CLU-conditional asset is *escalatability*, not a superior energy signal. [*evidence.*]
5. **The one-hop non-local edge, and its boundary (§4.2).** A direct wormhole edge has cost flat in the number of units where multi-hop diffusion scales (1.18×10^8 vs $1.76 \rightarrow 2.94 \times 10^8$ FLOP/query, $N \in \{4, 8\}$) with better distant accuracy — but *energy-gating* it **loses** to a 449-param physics-free router in FLOPs and accuracy, reported plainly as the boundary. [*evidence.*]
6. **The regime map, settled, and the corrected cost story (§4.3).** At matched accuracy a modern Hopfield network is the **cheaper and more cue-noise-robust** retriever (≈ 1 matvec); the CLU gate’s savings figure — $9.9/9.5/6.2 \times$ across kv32/64/96, falling with load — is *intra-CLU* rationing vs a full-budget CLU, never a cost win over Hopfield. On the full grid (198 jobs, $n = 8$) the gate reaches/reverses Hopfield **only on clean/correlated cues at kv ≤ 64** ($\Delta + 0.02$); beyond kv64 is an **epoch-budget wall, not a capacity wall**; and — the dominant negative — **no cell closes under cue noise** (gate 0.36 vs Hopfield 0.71 at $\sigma = 0.6/\text{kv32}$ despite fidelity ≈ 1.0). The gate rations *clean* retrieval; noise-robustness is Hopfield’s. [*evidence; cost final, accuracy regime-specific and settled.*]

The paper’s value is the certificate stack of §3 and the discipline of §4, not a leaderboard result. Everything is laptop-CPU. **We own the shape of this contribution profile explicitly:** the certificate stack is the contribution, and it is verified where we control the testbed; §4 maps precisely where it does and does not translate into a learned-memory advantage over a cheaper black box, and three of our six contributions are boundaries or negatives. §4 is the paper’s honest perimeter, not an incidental ablation.

2 Setup: a conservative memory and its reachable set

The map. One dissipative velocity-Verlet step with step ε and per-step damping $\gamma \in [0, 1)$ advances (q, p) by a kick–drift–kick of $H = T(p) + V_\theta(q)$ then $p \mapsto (1 - \gamma)p$. The update is **conformally symplectic**: $J^\top \Omega J = (1 - \gamma)\Omega$, $\det J = (1 - \gamma)^d$ in dimension d ; a position-gated

coupling scales this to $\det J = (1 - \gamma\varphi(q'))^d$. A **kinetic mode** fixes T : *relativistic* mode hard-caps $|\dot{q}_i| < c/\sqrt{M_i}$; *Newtonian* mode has $\dot{q}_i = p_i/M_i$ unbounded. A state-dependent **governor** $\gamma_n = s \tanh(\max(0, H - E^*))$ bleeds excess energy toward E^* . Full derivations and machine-precision checks are in the theory note.

Reachability, made falsifiable. For the governed map $\Phi_{\varepsilon, \gamma}$, the T -step **reachable set** from z_0 is $R_T(z_0) = \{\Phi^n(z_0) : 1 \leq n \leq T\}$, position shadow $Q_T(z_0)$. **Access** is provably enlarging Q_T . The theory note proves (Prop-A2) that in relativistic mode one drift advances q by at most $\varepsilon c/\sqrt{M_i}$ per coordinate, so

$$Q_T(z_0) \subseteq C_T(q_0) := \{q : |q_i - q_{0,i}| \leq L_i, L_i = T\varepsilon c/\sqrt{M_i}\}.$$

The **causal box** C_T is **energy-blind**: injected momentum drives $|\dot{q}_i|$ toward but never past $c/\sqrt{M_i}$. Hence the dichotomy:

- **REACH failure:** $q^* \notin C_T(q_0)$ (kinematic). Curable *only* by a nonlocal jump.
- **ESCAPE failure:** $q^* \in C_T(q_0)$ but trapped behind a barrier ΔV_b the dissipating budget cannot climb. Curable by any bounded injection faster than the governor re-brakes.

In Newtonian mode reach is instead energy-limited ($|\Delta q_i|_T \leq T\varepsilon\sqrt{2(H - V_{\min})/M_i}$), so energy *does* buy reach — which is why the safe headline mode (relativistic) is the mode where energy *cannot* buy *flow* reach (the cap bounds the Verlet drift regardless of budget), motivating a nonlocal mechanism. A squeeze can still buy reach *canonically* — as an instantaneous displacement, not through the flow — but only at the exponential energy price §3 makes explicit; that is the pricing law, not a loophole in the cap. The two failure modes need *separate* predictions (the disambiguation §3.3 returns to).

3 The certificate stack, verified

Reporting grade: verification. Every number in §3 is on a designed, architecturally-invariant analytic testbed (a symmetric double well along the reach coordinate; an $SO(2)$ sector for the latch), with **oracle channel placement**, dim 2 (headline) and 4 (identical), 5 seeds, laptop-CPU, $\gamma = 0$ rollout for a sharp box, mass band $[4.0, 0.25]$ ($16\times$ inertial contrast). These verify the theory's *exactness*; learned entrance-placement is out of scope (§5).

3.1 The two mechanisms and their receipts

Squeeze cures escape (bounded, re-absorbable injection). A mass-weighted Lorentz squeeze $S_\zeta^{(M)}$ is symplectic ($\det = 1$ exactly) and delivers both amplified velocity and displacement toward the saddle, $\delta q' = \delta q \cosh \zeta + p_0 \sinh \zeta$. Its injected energy is **bounded**, $H(S_\zeta z) \leq e^{2|\zeta|} H(z)$ (theory note Prop-12); the governor re-absorbs it in $\approx 2\zeta/\gamma_c$ steps. *Verified:* on the matched quadratic H the injection tracks the bound at every rapidity — $(\zeta, \text{ratio}, \text{bound}) = (0.25, 1.13, 1.65), (0.5, 1.55, 2.72), (1.0, 3.79, 7.39), (2.0, 27.5, 54.6)$ — and $\det S^{(M)} = 1.000 (\pm 4 \times 10^{-6})$. **Fine print (stated with the claim):** the $e^{2|\zeta|}$ bound is a *matched-quadratic- H* certificate; on the quartic well the raw ratio can exceed it (expected), so we quote it in that scope.

Wormhole cures reach ($\det J = 1$ exact, ledgered, latch transported). An intra-unit wormhole is a learned pair of loci (a, b) with a hard gate frozen at capture, applying the **constant canonical translation** $q \mapsto q + \Delta$, $p \mapsto p$, $\Delta = b - a$. Its Jacobian is I_{2d} , so $\det J = 1$ **exactly**,

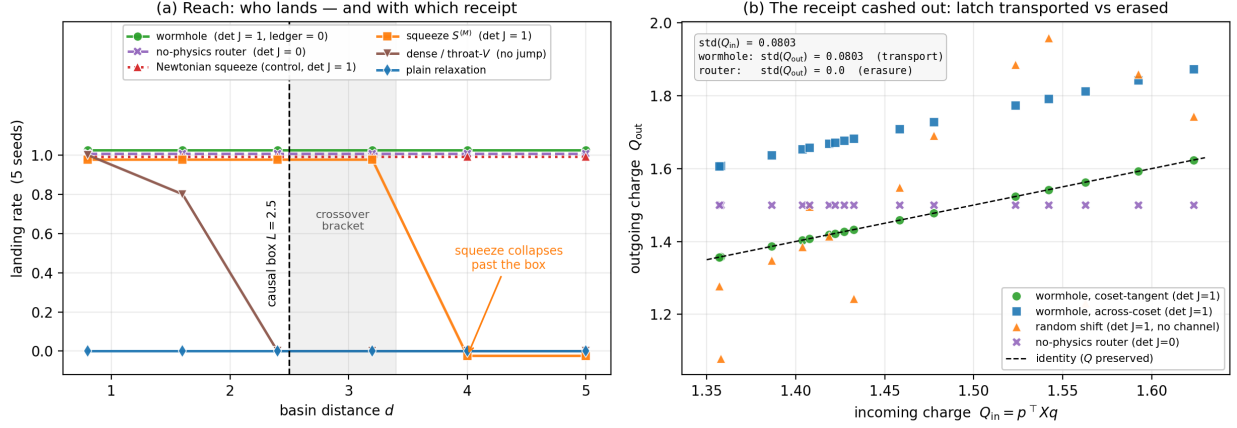


Figure 1: **Headline (§3, verification grade).** (a) **Reach:** landing rate vs. basin distance d on the designed double-well testbed (dim 2, 5 seeds, oracle placement, $L = 2.5$; *arms are offset vertically for visibility* — wormhole, state-replacing map and Newtonian control all land at exactly 1.0). Plain relaxation is escape-blocked everywhere; the squeeze $S^{(M)}$ steps up and its reach is then *priced out past the swept rapidity budget* $\zeta \leq 2.0$ (its reach is the bracket $[L, L + p_0 \sinh \zeta / M_0]$, shaded — the observed edge is $d \approx 3.2$, not a knife-edge at L ; the same bracket predicts $\zeta \approx 2.01$ to reach $d = 4.0$ and $\zeta \approx 2.64$ for $d = 5.0$, at exponentially rising energy — reach is *priced*, not capped); the wormhole is flat at all d with $\det J = 1$ and a *fixed* ledger = 0 exact; the Newtonian-squeeze control reaches past L , confirming the relativistic cap is the operative constraint on the flow; the dense/throat- V potential fails reach for $d \geq 2.4$. (b) **The receipt cashed out:** outgoing vs. incoming Goldstone charge for 16 states drawn in the capture ball. The $\det J = 1$ wormhole **transports** the charge (slope 1; exact constant shift $p^\top X \Delta$; spread preserved, $\text{std}(Q_{out}) = \text{std}(Q_{in}) = 0.0803$) while the untrained $\det J = 0$ **state-replacing** map **erases** it (slope 0; all 16 states exit at one common charge, $\text{std}(Q_{out}) = 0$). The fine print is in the figure too: a $\det J = 1$ *random shift* also scrambles Q — volume alone is not the latch receipt; the matched channel is.

independent of Δ and the gate (the flow of the linear Hamiltonian $W = \Delta \cdot p / \tau$). Its only cost is a discrete jump $\Delta H_{wh} = V_\theta(q + \Delta) - V_\theta(q)$ that must be **ledgered**; matched loci give free transport. *Verified:* symmetric double well, channel $-a \rightarrow +a$, ledger = 0.0 exact, state teleports to $q = +1.000$; $\det J = 1.000$. **Fine print:** a gate varying *during* the jump gives $\det J = 1 + \nabla g \cdot \Delta \neq 1$ (unit test 2.05) — an unpaid contraction; the receipt is clean only for a frozen gate (a proven design guard, App. D N31).

Latch transport ($p^\top X \Delta$, exact). For a protected direction carrying Goldstone charge $Q = p^\top X q$ (the theory note’s coset content, X the broken generator), under the wormhole $Q' - Q = p^\top X \Delta$ *exactly*. The wormhole moves *one* phase point, so it **transports** the latch (not copy, not erase): preserved iff $\Delta \perp X^\top p$, else shifted by the exact $p^\top X \Delta$. *Verified:* zero-shift $\Delta Q = 0.0$ (pred. 7.5×10^{-10}); across-coset $\Delta Q = 0.2500 = p^\top X \Delta$; raw squeeze preserves Q ($\Delta Q = 1.2 \times 10^{-7}$); random shift erases ($\Delta Q \in \{0.035, 0.157, -0.143, 0.302, -0.144\}$). The receipt distinguishes *transport* from *erase* — the certificate that a paid jump carries memory. **Fine print, stated with the claim (C-6):** $\det J = 1$ *alone* is not the latch receipt — the random-shift baseline is itself volume-preserving and still scrambles Q ; what buys transport is the **matched channel** (Δ coset-tangent, $X \Delta \perp p$) together with the ledger.

BIBO fine print, stated with the wormhole claim (C-6). $\det J = 1$ certifies *volume*, not *boundedness*. An exit placed outside the coercive connected component of V_θ can escape to infinity even though the jump is symplectic and its ledger bounded — and, sharper, *even when the ledger is free*. On a controlled non-coercive well ($V = \frac{1}{2}kq_0^2 - \epsilon q_0^4$, coercive edge $x_b = 3.536$) the exit at $b = 5.0$ has $\Delta H = 0.0$ *exactly* — cheaper than the admissible $b = 3.0$ exit’s $\Delta H = 2.88$ — and an energy-only sub-level test **admits** it, yet the trajectory escapes with $r^* \propto T$ (growth $r^*(2T)/r^*(T) = 1.91$), while the screened exit stays bounded (ratio 1.000). **Coercive-component membership, not the energy ledger, is the operative BIBO clause** (6/6 exits predicted; App. B.2, Fig. 4). Full receipt: Table 1 (App. B).

3.2 The discriminating experiment: squeeze reach is priced, the wormhole is flat-priced — and only the wormhole hands back a receipt

The falsifiable heart is a single controlled reach battery: a K -basin double well with basins spanning below/above the box $L = 2.5$ ($T = 100$, $\varepsilon = 0.05$, $c = 1$, heavy reach coord $M_0 = 4.0$), the governor fixed so plain relaxation *provably* cannot leave the start basin (escape-blocked, $\text{KE}_0 = 0.72 < \Delta V_b = 1$). Figure 1 is the headline: panel (a) the landing rates, panel (b) the receipt that separates the arms landing rates cannot. Landing rate vs. distance d (dim 2, 5 seeds; $< L = \{0.8, 1.6, 2.4\}$, $> L = \{3.2, 4.0, 5.0\}$; dim 4 identical):

arm	$d < L$	$d > L$	$\det J$ (measured)	reading
plain relaxation	0	0	$(1-\gamma)^d$	escape-blocked everywhere
squeeze $S^{(M)}$	1	$\rightarrow 0^\dagger$	$1.000 \pm 4\text{e-}6$	steps up; reach beyond the box is priced
wormhole	1	1	1.0 exact, ledger 0.0	flat ≈ 1 at all d , with a receipt
Newtonian-squeeze (control)	1	1	1.0	energy buys flow reach — confirms the cap
state-replacing map (no-physics)	1	1	0.0 exact	reaches by <i>overwriting</i> , volume-annihilating
dense/throat- V	$\rightarrow 0$	0	$(1-\gamma)^d$	helps near, fails reach

[†] The squeeze’s $\rightarrow 0$ at $d \in \{4.0, 5.0\}$ means “priced out of the swept $\zeta \leq 2.0$ grid,” *not* “cannot reach”: the verified bracket predicts $d = 4.0$ lands at $\zeta \geq 2.0105$ ($\approx e^{4.02}H$) and $d = 5.0$ at $\zeta \geq 2.6441$ ($\approx e^{5.29}H$) — analytic predictions from the bracket, not new runs; the grid ended at $\zeta = 2.0$. **Two “router” objects:** the **no-physics.router** arm here is an *untrained analytic constant map* (a state-replacing baseline), distinct from §4.2’s *learned decision head*, which routes *through* the wormhole’s $\det J = 1$ edge and inherits its receipt.

Figure 1 is deliberately two-panel because the landing-rate axis *cannot* display this paper’s thesis: the wormhole, the state-replacing map and the Newtonian control all land at 1.0 and coincide. What distinguishes them is the receipt column — and panel (b) is that column made visible.

What each arm certifies. (i) The **squeeze prices reach; it does not fail at the box**. It is a canonical map, not a flow: applied at the basin bottom it grants an *instantaneous* displacement $(p_0/M_0) \sinh \zeta$ before the capped flow adds at most L , so its reach is the bracket $[L, L + p_0 \sinh \zeta / M_0]$ — it *does* exceed the box, by a ζ -controlled amount (observed edge $d \approx 3.2$, not a knife-edge at $L = 2.5$). But that excess costs energy $\leq e^{2|\zeta|}H$ (§3.1), so reach beyond L carries an **exponential price in rapidity**: the swept $\zeta \leq 2.0$ reaches $d \lesssim 3.6$ ($\leq e^4 \approx 55H$), and the same verified bracket predicts $d = 4.0$ needs $\zeta \approx 2.01$, $d = 5.0$ needs $\zeta \approx 2.64$. The falsifiable content, **sharpened from a collapse into a pricing law**: *reach via squeeze is exponentially priced in distance; reach via wormhole is flat-priced*. The squeeze’s $\rightarrow 0$ entries mean “priced out of the swept grid,” not “cannot reach.” (ii) The **wormhole** is flat at all d with $\det J = 1$ and a *fixed* ledger = 0 per jump, independent of d . (iii) The **Newtonian control** reaches past L (energy buys *flow* reach uncapped) — the control showing the relativistic cap, not a coding artifact, is the constraint on the flow. (iv)

The **state-replacing map** (the untrained no-physics baseline) matches the wormhole’s landing but by fiat, by *overwriting* the state. Its map $(q, p) \mapsto (b, p)$ has Jacobian $\text{blockdiag}(0_d, I_d)$, so $\det J = 0$ **exactly** (measured by forward-mode autodiff): *volume-annihilating and non-invertible*, a strictly stronger statement than “carries no certificate,” cashed out in §3.2.1. **It is not §4.2’s learned router**, a decision head that routes *through* the wormhole’s $\det J = 1$ edge (decision and transport are orthogonal). (v) The **dense/throat- V** arm fails reach for $d \geq 2.4$: a nonlocal potential lowers the barrier but the trajectory still traverses it, staying bound by C_T . The wormhole is the *only* arm reaching all d **at a fixed ledger with a $\det J = 1$ receipt**.

3.2.1 The receipt cashed out: a state-replacing jump erases the latch a canonical translation transports

A certificate that never changes an outcome is decoration. Here it changes one, measured on the same designed testbed (Fig. 1b; dim 2 headline, replicated at dim 4).

Decision is not transport — state this once, sharply. Two objects in this paper are loosely called “router,” with opposite transport semantics. §4.2’s **router_mlp** is a *learned decision head*: it *decides whether* a query takes the non-local edge and transports *through* the wormhole’s own $\det J = 1$ channel (“routes via the **same** direct wormhole edge”), inheriting the receipt and erasing nothing. The object here is different in kind — an *untrained analytic map* that *is* the transport, and a **state-replacing** one: it overwrites $(q, p) \mapsto (b, p)$, annihilating phase volume. **A learned gate bolted onto a certified channel is not a counterexample to the receipt; it is a consumer of it.** The certificate prices only transport, never the decision.

We draw 16 incoming states uniformly in the capture ball (radius $0.3 < \rho = 0.35$) around an entrance on the vacuum circle at fixed momentum p , and read $Q = p^\top X q$ for each; the incoming cloud has $\text{std}(Q_{\text{in}}) = 0.0803$.

arm	$\det J$	ΔQ mean	ΔQ std	pred. $p^\top X \Delta$	max err	$\text{std}(Q_{\text{out}})$
wormhole, coset-tangent	1.0000	−0.0000	0.0000	0.0000	1.2e^{-7}	0.0803
wormhole, across-coset	1.0000	0.2500	0.0000	0.2500	0.0	0.0803
random shift (no channel)	1.0000	0.0972	0.2465	(none)	—	0.2793
state-replacing map	0.0000	0.0379	0.0803	(none)	—	0.0

The guarantee. The wormhole’s canonical translation is *injective*, so it shifts **every** incoming state’s charge by the **same exact constant** $p^\top X \Delta$ (ΔQ std = 0, err $\leq 1.2 \times 10^{-7}$). The stored spread survives: $\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}})$ to all printed digits, and the pre-jump state is exactly recoverable, $q_{\text{in}} = q_{\text{out}} - \Delta$ (max err 2.2×10^{-8}).

The violation. The state-replacing map sends the whole capture ball onto one point: all 16 states exit at a single common charge (max $|Q_{\text{out}} - Q_{\text{out}}[0]| = 0.0$, bit-identical) and $\text{std}(Q_{\text{out}}) = 0$ **exactly** — the latched coset content is *irrecoverable*. This is $\det J = 0$ cashed out as a measured downstream consequence, replicating in every cell (dim $\in \{2, 4\} \times \text{seed} \in \{0, 7\}$: $\text{std}(Q_{\text{out}})_{\text{wh}} = \text{std}(Q_{\text{in}})$ in all four — 0.0803/0.0679/0.0533/0.0448 — and $\text{std}(Q_{\text{out}})_{\text{map}} = 0$ in all four).

The claim at its honest altitude. The receipt separates **transport** from **state-replacement**, with a measured consequence: at $\det J = 1$ the matched canonical channel *transports* the stored charge; a $\det J = 0$ state-replacing jump *erases* it. The claim is about *maps*, not ideologies or learnedness — which makes it robust. Three qualifications travel with that sentence. (i) **Volume alone is not the latch receipt** — the random-shift arm has $\det J = 1$ and still scrambles Q (out-spread 0.2793, $3.5 \times$ the incoming spread); the *matched channel* preserves it. (ii) This is a **mechanism-level violation on a designed testbed with oracle placement**, not a learned-system win, and the erased object is the *untrained state-replacing baseline*, not §4.2’s learned router.

§4.2’s `router_mlp` is a **decision head**: it beats the energy-gated edge in FLOPs and accuracy at *choosing* whether to take the non-local edge, by transporting *through* the wormhole’s own $\det J = 1$ channel — inheriting the receipt, erasing nothing. Decision and transport are orthogonal; both facts hold. (iii) For the BIBO half (App. B.2) the attribution is narrower: what buys boundedness is **the receipt, not the jump**, since an unscreened wormhole and the state-replacing map coincide exactly.

3.3 Why a prior null does not bear on this claim

A memory of this class was previously reported to gain nothing from squeeze/boost *retries* at single-unit scale (a pooled null; App. D N1). That null tested **selection among already-reachable** attractors (retrieving the wrong pattern) at near-uniform mass ($S^{(M)} \approx S$) with within-basin perturbations — none of its targets sat *outside* the reachable set. The reach claim asks a categorically different question — land in a basin *provably outside* plain relaxation’s reach, with a $16\times$ mass contrast — verified by the §3.2 crossover. We flag this to preempt “didn’t your own retries fail?”: the two share only the operator $S^{(M)}$.

4 Three honest pillars on learned memories

Reporting grade: evidence. §4 leaves the designed testbed for **trained** memories; each claim is scoped and priced in-line. Where the receipt says a cheaper black box wins, we say so. **The scoreboard sentence, stated up front: external benchmarks won on their own headline metric = ZERO.**

4.1 Calibrated compute-rationing — memory-agnostic mechanism, escalatable-memory asset

We equip a trained CLU associative memory (MQAR key-value recall, vocab 256, kv $\in\{16, 24, 32\}$, 5 seeds, laptop) with a self-calibrating gate: at write time it runs a jittered-cue self-test and fits a per-model calibration head (Platt 1999) mapping a relaxation residual to p_{wrong} , plus a learned exit threshold by **Learn-then-Test** (LTT; Angelopoulos et al. 2021). Deployment queries run a staged-relaxation ladder and exit when calibrated-confident. Three findings:

1. **Calibration transfers, and the mechanism is memory-agnostic.** Raw residual energy is not cross-model comparable (pooled AUROC 0.431 ± 0.038 , anti-ranked); the per-model head makes it deployable (0.869 ± 0.015 , 5 seeds). The *identical* stack on a modern Hopfield memory (Ramsauer et al. 2021) reproduces the jump (raw 0.18 $\rightarrow$ calibrated 0.88, matching CLU 0.43 $\rightarrow$ 0.87) — the gate mechanism is **not** CLU-specific. The value is the calibrated-rationing apparatus, not a claim that CLU energy is a better confidence signal.
2. **The CLU-conditional asset is escalatability — and the allocation payoff is largest at the easy band.** What does not transfer to a one-shot memory is the allocation payoff: on the CLU’s graded relaxation the learned point spends $4.81 \pm 0.44\times$ fewer steps **at kv16, at a 400-epoch budget** (0.894 ± 0.021 @ 629 ± 60 vs always-full 0.847 ± 0.037 @ 3000) while landing *above* always-full accuracy. **The epoch budget is load-bearing and we own it inline (C-7):** §4.1’s models are trained 400 ep, a band §4.3 shows is not yet converged. At convergence (§4.3, App. C.4.a: 2000 ep) the gate’s accuracy **matches** full-budget rather than exceeding it — the robust payoff is *rationing* ($9.9\times$ intra-CLU at kv32/2000 ep — §4.3’s grid, a different epoch budget from this subsection’s), and the accuracy headroom is a property of

an imperfect (under-trained) memory, not of the mechanism; CM-2’s “escalatable memory” claim rests on the saving. **The magnitude of the saving also does not persist as the band hardens, and we name the levels rather than the band:** kv24 gives $1.57 \pm 0.07 \times (0.547 \pm 0.039 @ 1919 \pm 82, +2.2 \text{ pts over always-full})$ and kv32 gives $1.14 \pm 0.06 \times (0.286 \pm 0.037 @ 2636 \pm 127, +0.6 \text{ pts})$. What survives at all three levels is the weaker, cleaner invariant: **the learned gate never pays full price for less accuracy at any level**, and savings scale with how much of the workload is confidently easy — a $4.8\times$ headline is a kv-16 number, not a property of the mechanism. A one-shot memory earns no such payoff at *any* level. Scoped to MQAR vocab-256, $kv \in \{16, 24, 32\}$, 5 seeds, laptop.

3. **Distribution-free certificates, assumption stated.** The LTT wrapper was valid on 30/30 cells (zero violations), e.g. coverage 0.647 ± 0.063 at target risk $\varepsilon = 0.05$ (measured risk 0.030), with graceful refusal where the memory is too weak. **Fine print, next to the claim:** LTT rests on an *exchangeability* assumption between write-time probes and deployment queries, which our jittered protocol only approximates — the calibration is *under-confident* (ECE $\approx 0.100 \pm 0.021$, safe side but a real shift), so the guarantee holds “within the stated probe-to-deployment scope.”

Two negatives on the record (App. D). (a) The gate does not let this memory out-abstain a near-perfect Hopfield ($0.983\text{--}1.000$ accuracy $\Rightarrow$ selective-risk ≈ 0 ; N2). (b) Residual energy adds essentially nothing over the readout margin ($\Delta\text{AUROC} \in [-0.004, +0.024]$); energy-as-a-superior-signal is a claim we **do not make** anywhere (N3). The pillar is the certified rationing apparatus on an escalatable memory.

4.2 The one-hop non-local edge, and where energy-gating it loses

A conservative memory can be given a **non-local edge**: a gated wormhole coupling transporting a query key to a distant archive, an attention-like long-range access. The surviving mechanism claim is about *cost structure*: the **direct one-hop edge has cost flat in N** (1.18×10^8 FLOP/query at $N \in \{4, 8\}$) where **N -hop chain diffusion scales** ($1.76 \times 10^8 \rightarrow 2.94 \times 10^8$) *and* loses distant accuracy ($0.41 \rightarrow 0.28$) — a direct edge beats hop-by-hop diffusion, in FLOPs and accuracy (5 seeds, MQAR-style, $N \leq 8$, laptop).

The honest boundary, stated plainly as the section’s spine: **energy-gating that edge loses**. A parameter-matched (449-param, 2-layer) physics-free router on the raw cue beats the energy-gated wormhole in *both* FLOPs and accuracy — router $1.000/0.948$ (local/distant, $N = 4/8$) @ 8.81×10^7 FLOP vs gated $0.887/0.715$ @ 1.18×10^8 — across all mixes $\{50/50, 80/20, 95/5\}$ and both N , over 5 seeds. Near-linearly-separable per-unit key clusters let the cheap classifier learn local-vs-distant trivially while the energy gate rediscovers the partition via relaxation, noisier. **The direct edge is the mechanism; energy is not the routing signal.** (Scope caveat, inline: linearly-separable cues drive the router’s dominance; a harder band is the untested fair stress test; App. D N24/N27.)

4.3 The regime map and the corrected cost story

Cost (final). A modern Hopfield network reaches its ceiling in ≈ 1 matvec ($0.947\text{--}0.979$ at $\beta \geq 5$; extra iterations change accuracy ≤ 0.003), so its floor is $O(kv \cdot d)$. **At matched accuracy, Hopfield is the cheaper retriever.** The CLU gate’s savings — $9.9/9.5/6.2\times$ across kv32/64/96 (App. C.4.a), a curve that falls with load, not a constant — is an **intra-CLU** number — gate vs full-budget CLU (3000 Verlet steps) — a compute-rationing measurement, **not** a cost win over Hopfield. “Matching Hopfield” means the gate’s accuracy *reaches* Hopfield’s, not that it does so more cheaply. We state this next to every accuracy-improvement claim.

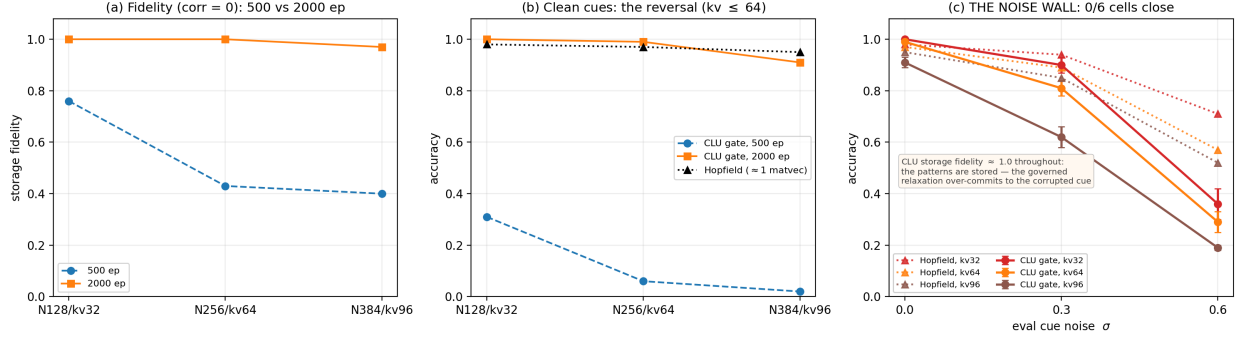


Figure 2: **The settled regime map (§4.3, evidence; regime-remap-2000ep, 198 jobs).** (a) Storage fidelity on the paired capacity axis, 500 vs 2000 ep ($n=8$): the 500-ep “Hopfield-dominant” map was an under-training artifact. (b) Clean cues: at 2000 ep the gate reaches/reverses Hopfield for $kv \leq 64$ ($\Delta + 0.02$) at $9.9/9.5/6.2 \times$ *intra-CLU* rationing across kv32/64/96 (never a cost win over Hopfield, which answers in ≈ 1 matvec). (c) **THE NOISE WALL — the dominant negative, plotted.** Under eval cue noise $\sigma \in \{0.3, 0.6\}$ the gate (solid) falls below Hopfield (dotted) at every capacity — 0/6 cells close — *despite* CLU storage fidelity ≈ 1.0 : the patterns are stored, but the governed relaxation over-commits to the corrupted cue. Error bars are ± 1 s.d. over seeds for the gate ($n=5$ at $\sigma > 0$; $n=8$ at the $\sigma=0$ clean reference, taken from the $\text{corr}=0$ capacity axis); the source reports no per-seed spread for the stress-axis Hopfield arm, so its curves carry no bars.

Accuracy (settled — full grid, 198 jobs). The initial “Hopfield-dominant 26/26” map was an **under-training artifact** (500 ep: gate 0.02–0.31 vs Hopfield 0.95–0.98, confirmed $n = 8$). Trained to convergence (2000 ep) CLU fidelity rises to ≈ 1.0 and the accuracy reversal appears **but is regime-specific, not general**, and never touches the cost story above (every savings number here — $9.9/9.5/6.2 \times$ at kv32/64/96 — is *intra-CLU* rationing, not a win over Hopfield). Three qualifiers travel together. (1) **Clean/correlated cues, $kv \leq 64$: the gate reverses Hopfield** (0.99–1.00 vs 0.97–0.98, $\Delta + 0.02$, $n = 8$); at $\rho = 0.9$ the reversal *widens* ($\Delta + 0.08$ to $+0.16$) **only because Hopfield collapses on correlated keys** (0.72/0.59) while the CLU gate *also* drops (0.87/0.67) — Hopfield fragility, not CLU strength. (2) **Beyond kv64 is an epoch-budget wall, not a capacity wall**: at 2000 ep a wall is visible (gate 0.99 $\rightarrow$ 0.92 $\rightarrow$ 0.60 across kv64/96/128) but at 4000 ep kv96 reverses (0.975 vs 0.947, $\Delta + 0.03$) and kv128 only ties ($\Delta + 0.004$); required epochs scale with kv, and small cells **over-train** (kv32 gate 1.00 $\rightarrow$ 0.89, 2000 $\rightarrow$ 4000 ep). (3) **THE NOISE WALL (dominant negative, foregrounded)**: under cue noise $\sigma \in \{0.3, 0.6\}$ **no cell closes at any capacity, even kv32** (gate 0.36 vs Hopfield 0.71 at $\sigma=0.6$ /kv32; Δ from -0.05 to -0.35), *despite* CLU fidelity ≈ 1.0 — the relaxation gate is markedly less noise-robust than one matvec, on-thesis for the position (*the gate rations clean retrieval; noise-robustness is Hopfield’s*). **Tally at 2000 ep: 6/15 cells close** (all correlation-axis, $kv \leq 64$; eval-noise axis 0/6). Figure 2 is the honest regime map — (a) fidelity, (b) the clean-cue reversal, and (c) **the noise wall itself**, plotted rather than merely asserted; Figure 3 the epoch-scaling frontier; full tables in App. C.4. **Net: the gate reaches or exceeds Hopfield accuracy only for clean/correlated cues at $kv \leq 64$** (kv-scaled epochs extend this on the clean axis); Hopfield keeps both the cost and the cue-noise-robustness advantage; the CLU asset is escalatable accuracy under a rationing gate on clean retrieval, not a general accuracy or cost win. Separately, the anchor training intervention that rescues a designed vacuum does *not* transfer to memory fidelity (it pins a structureless init; App. D N30) — longer training, not the anchor, closes the gap.

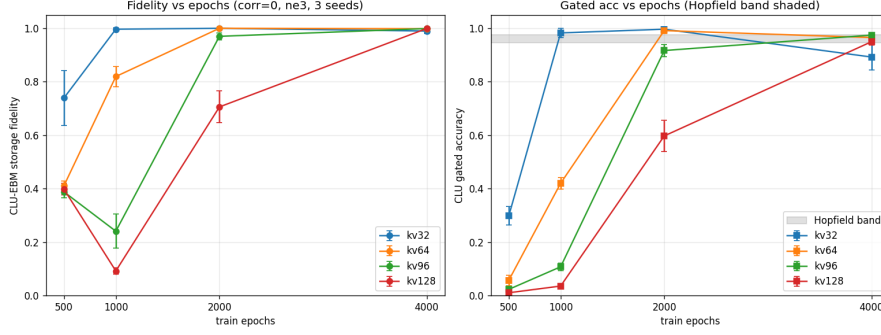


Figure 3: **Epoch-scaling frontier (§4.3, evidence)**. Gate accuracy and storage fidelity vs training epochs across $kv \in \{32, 64, 96, 128\}$, against the epoch-independent Hopfield band (0.947–0.976): an epoch-budget ridge, not a capacity wall; kv32 over-trains ($1.00 \rightarrow 0.89$). (*Candidate for the appendix at the pruning pass.*)

4.4 A matched-compute anytime read (beyond the three pillars): it ties

*Reporting grade: evidence, with the substrate stated — a **designed** store addressed through a **learned, frozen** encoder φ , not a trained CLU checkpoint. Every number in this subsection is 6 seeds of its own protocol, never §3/§4.1–4.3’s 5 seeds.*

The three pillars above ration or route *within* a memory. The comparison a reviewer asks for next is against a floor handed the *same* compute budget, and we report it here. In the native regime — pixel-space corruption of a φ -addressed store, where **no mask oracle can be constructed** — the confidence-gated anytime read **ties** the matched-compute feedforward-in- φ floor ($+0.8 \pm 1.6$ pp, **6 seeds**, 3/6 positive) and beats 1-shot kNN-in- φ by $+1.1 \pm 1.4$, while beating every mechanism control: $+4.6 \pm 2.2$ pp over an equal-energy random kick, 6/6 seeds, $+4.4$ over an ensemble of restarts, ungated retry collapsing to 0.000, and auto-stopping at $1.40 \pm 0.20 \times$ compute. **It ties; it does not win** — and the tie is the result, because it is the first cell we have measured in which the anytime read is not *below* the correct floor. It is also the measured object behind §5’s proof-of-concept framing: the deferred matched-compute comparison, run in one regime, comes back level.

Scope, in-sentence (C-5). One dataset (MNIST-class images, φ fit on the first task’s classes only and frozen), one corruption level (pixel dropout $p = 0.8$), one crowded store (199–200 wells, end-of-stream), 6 seeds, laptop-CPU. At the easier level ($p = 0.5$) the cell is **saturated** — first pass 0.99, floor 1.000 — and every line is flat; on CIFAR-class images the 1-shot kNN-in- φ floor is 1.000 and the same read sits *below* it (-0.055 ± 0.019). A saturated floor is unbeatable, which is why the hard cell is the one that carries the result, and why nothing here generalises past it.

The negative that travels with any anytime claim (C-9). Where a mask oracle *can* be constructed, the verdict is a **decision-grade NO with headroom present**, so it cannot be blamed on saturation: built from ambiguity (block occlusion plus a contracted crowded store, 3 seeds, first pass 0.26–0.93 off ceiling), the gated read does **not** beat the ML-optimal masked-erasure oracle ($= 1.000$) in *any* of 8 cells (-3.5 – -38 pp). The corrected status is precise: the **static-retrieval instantiation** of the compute-adaptive-read claim is **closed** by that measurement, while the **mechanism** survives all of its controls (kick and ensemble flat or declining, directed boost $+2.3$ – $+36.2$ pp, auto-stopping $\times 1.20$ – 1.81) — losing to an oracle that is *told which coordinates were erased* is the metric-native ceiling, not a falsification of the dial. On any erasure protocol

the mandatory baseline is that masked observed-dims nearest-neighbour oracle, never the naïve full-vector one.

Quoting an anytime curve, including a flat one (C-6). A flat anytime curve does *not* license “the store carries nothing.” Flatness licenses only the disjunction “*the store carries nothing or it cannot be addressed,*” and the two are separable by a control we ran: same store, same physics, same frozen budget grid, the read moves $0.0223 \rightarrow 0.8219 \rightarrow 0.8711 \rightarrow$ plateau under oracle addressing where the shipped read is flat at 0.0004 — quoted as the curve it is, never as an endpoint pair. And **no uniqueness is claimed for the anytime shape**: that figure is an occupied venue (deep-equilibrium models, energy-based transformers, recurrent-depth architectures); what is claimed is a shape-and-mechanism result measured against its own controls.

5 Position, scope, and horizon

The position, restated. On a conservative memory, test-time compute is *paid access*, and the price list is physical: a squeeze buys escape for a bounded, re-absorbable injection and **prices reach exponentially** (it exceeds the box only by its instantaneous displacement $p_0 \sinh \zeta / M_0$, at energy cost $e^{2\zeta}$); a wormhole buys **unbounded** reach for a $\det J = 1$ receipt and a *fixed* discrete ledger, and transports rather than destroys stored content — where an untrained $\det J = 0$ **state-replacing** jump, which reaches by overwriting the state, provably erases it. (Deciding *whether* to take that certified edge is separate: §4.2’s learned decision head routes *through* the channel and inherits its receipt — decision and transport are orthogonal.) These are theorems verified to machine precision (§3), composing with the memory’s BIBO guarantees *only under coercive-component screening* — a bounded, or even free, energy ledger is not enough (App. B.2). Where a learned halting head or an energy verifier (Gladstone et al. 2025) escalates *the same operation* with no structural account, a symplectic memory prices each access mechanism *and certifies what it did to the state*. The honest pillars of §4 delimit what the extra compute does *on this memory*: a rationed relaxation budget buys accuracy on the memory we trained (§4.1, §4.3), and we offer that as an **empirical proof of concept**, not as a comparative result — **no matched-compute floor and no matched-bytes exemplar store is run against §4.1–4.3 anywhere in this paper**, and that comparison is named here as **future work** (the one matched-compute comparison we do report, §4.4, comes back a *tie*). They also delimit where a cheaper black box wins (routing): a certificate stack is a design discipline, not a guarantee of dominance. **The position has a third corner, and it is an impossibility result.** Beyond a single access mechanism, the same accounting binds the store’s *read*: with a quadratic payload channel, the $q_2(0) = 0$ anti-decoration guard and a fixed read budget, a store cannot have all three of **exact value fidelity**, **amplitude-independent address hold** and **amplitude-independent read latency** — pick two. The shipped store drops the second, which is why effective lifetime correlates $r = -0.85$ with a_i^2 ; the recommended gated-stiffness channel drops the third. **Both proposed fixes to that corner are refuted, and neither may be described as available.** The corner to lead with is the third: *dropping amplitude-independent latency is the compute-adaptive-read dial — a faded memory costs more integration steps to read, which is a physical, measurable statement a timestamped row cannot make.* (Proved and measured on this program’s decaying-store instantiation — a designed store with a quadratic payload channel and a fixed read budget, 3 seeds, laptop-CPU — not on the §3 unit.)

Scope (stated, not buried). The §3 verifications use **oracle channel placement**, dim 2/4, 5 seeds, laptop-CPU; the §4 learned-memory pillars are MQAR-style, vocab-256, kv/ N small, laptop-

CPU. No claim here is at scale, and none uses learned placement. And the substrate scope, stated once, in our own voice: **these laws govern the store; end-to-end performance additionally depends on the encoder, measured separately, φ -bytes ledgered.**

Horizon / future work.

- **Learned entrance-steering is the engineering crux.** The reach theorem proves the outer reachable set, but a trajectory must still *arrive* at a channel entrance under the flow; placing/learning (a, b) so entrances sit on natural trajectories is the true cost at scale and the likely failure point — addressed in forthcoming work.
- **Certifying exits on a *learned* potential.** The BIBO receipt (App. B.2) screens exits against the coercive component of an *analytic* well with closed-form x_b . For a learned, architecturally non-coercive V_θ the boundary is unknown; a sub-level-set estimator (or restoring the $\alpha\|q\|^2$ confinement Deep/Conv drop) is the natural next experiment. Our coercive screen is an oracle, exactly as our channel placement is.
- **γ -re-absorption timing.** The governor re-absorption certificate ($t_{\text{reabsorb}} \approx 2\zeta/\gamma_c$) is verified only to leading order at $\gamma = 0$; a $\gamma > 0$ sweep closes that row.
- **Retry acceptance as a certified kernel** is developed as the closing design-rules below (derivations in App. F).

Test-time retries as a certified Markov kernel — four design rules (theory-complete on toy EBMs; rule 2’s mechanism controls are measured on a designed store; no runs on trained CLU checkpoints are claimed). The paid-access logic extends one level up, from a single access mechanism to the *acceptance rule* governing repeated retries. Because the squeeze $S_\zeta^{(M)}$ is exactly symplectic ($\det J = 1$), a sign-symmetrized squeeze accepted by $\min(1, e^{-\Delta H/T})$ needs **no Jacobian correction** and is a π -reversible kernel for the Gibbs measure of the trained energy — test-time compute as MCMC with a *stationarity certificate* (Duane et al. 1987; Neal 2011). Four rules follow, each with its receipt, and each a restriction rather than a promise.

1. **Run certified segments at $\gamma = 0$; keep the governor outside them.** State-dependent $\gamma(H) > 0$ is non- π -preserving dissipation, so the actual squeeze-then-relax cascade is **Metropolis-within-annealing** toward a colder, MAP-seeking measure ($T_{\text{eff}} : 1.0 \rightarrow 0.61$ as $\gamma : 0 \rightarrow 0.2$, verified) — possibly *desirable for retrieval*, but **not Gibbs sampling**. **We never claim stationarity for the governed composite.**
2. **Mix; do not rely on the squeeze alone.** The squeeze family is a one-parameter subgroup whose orbit is a single hyperbola, hence **reducible/non-ergodic**; read it as a mass-metric-preconditioned *reach* move and use $\frac{1}{2}$ MALA(σ^*) + $\frac{1}{2}$ sign-symmetrized squeeze-MH (a mixture of π -reversible kernels is π -reversible). **Measured, on a designed store (evidence; MNIST, designed/closed-form on every line, an 8-cell ladder at seed 0 re-measured across 5 seeds of that protocol — never §3/§4’s 5 seeds):** the lift is the **directed** symplectic re-launch: equal-energy random kicks and ensembles of k independent restarts are **dead flat in all 8 cells**, while confidence-gated CLU retry rises **+6.6+76.2 pp** and **auto-stops at $\times 1.2$ – 1.8 compute**; the confidence gate is load-bearing — ungated retry-all collapses $0.96 \rightarrow 0.004$ at $9\times$ compute. Three companions travel with that sentence. *(i) The floor:* **CLU draws a rising, auto-stopping accuracy-vs-compute curve that a saturated feedforward memory structurally cannot draw, while remaining below that saturated floor on masked-pixel retrieval** — on 5 seeds of the protocol the gap

to that floor is -3.9 – -20.7 pp under masking and -9.7 – -48.2 pp under noise, negative in 40/40 (seed, cell) pairs, with gated lift $+0$ – $+79.7$ pp and 0/40 violations by either mechanism control. (ii) *The protocol split*: the boost recovers mask queries strongly ($+36$ – $+76$ pp) and Gaussian-noise queries weakly ($+6.6$ – $+34.8$ pp), and **nothing recovers past $\sigma \approx 0.4$** . (iii) *The acceptance discipline*: cosine-to-nearest-well is a *ranking*, never an acceptance, signal. **And the same decision-grade NO with headroom present that §4.4 states applies here**: against the ML-optimal masked-erasure oracle ($= 1.000$) the gated read does **not** win in any of 8 cells (-3.5 – -38 pp, 3 seeds, first pass 0.26–0.93 off ceiling), so this is a mechanism claim and never a leaderboard claim.

3. **Calibrate the Langevin noise by fluctuation–dissipation (Newtonian kinetic mode).** The discrete FDT scale $\sigma_i^* = \sqrt{M_{\text{eff},i} T \gamma (2 - \gamma)}$ is load-bearing, not cosmetic: it moves the sampler from shadow-biased to exact (L_1 0.0995 $\rightarrow$ 0.0065, float64 toy EBMs). *Scope (a scope clause, not a caveat — V1’s units are Newtonian throughout)*: this exactness is a Newtonian-mode statement. Once the step is Metropolis-adjusted, σ^* is a *proposal-tuning* scale (mixing efficiency), not a correctness condition — any σ leaves π invariant under MH, and the 0.0995 $\rightarrow$ 0.0065 gain is the finite-budget shadow bias it removes. In a relativistic kinetic mode the momentum target is Maxwell–Jüttner, for which a fixed-covariance Gaussian kick is not a Gibbs refresh, so there the MH correction — not σ^* — secures stationarity (App. F.2).
4. **Project every proposal off the coset tangent — even the certified retry carries a receipt.** The Gibbs measure is *exactly flat* along the Goldstone coset, so coset-tangent proposals are accepted w.p. 1 and **random-walk the stored register** at $D = \frac{1}{2}s^2$, erasing it after $N_{\text{erode}} \approx (\Delta_{\text{read}}/s)^2$ accepted moves [proven; verified $D = 1.29 \times 10^{-3}$ vs $\frac{1}{2}s^2 = 1.25 \times 10^{-3}$] — true even for a *charge*-preserving isotropic squeeze, which conserves Q while eroding coset *position*. Projection quenches it ($D = 0$, verified).

Per the energy-is-not-a-signal discipline held throughout: the acceptance temperature that makes the kernel *useful* is a per-model, write-time-calibrated quantity — the same learned gate §4.1 already fits — so **the value of the MH framing is the stationarity certificate and the explicit erosion accounting, not parameter parsimony, and not, on current evidence, performance**. Derivations, the four numerical checks, and the specified discriminating experiment are in App. F.

Prior-art honesty. The nonlocal edge overlaps attention, skip connections, and MoE routing (Shazeer et al. 2017; Raposo et al. 2024); the squeeze overlaps simulated-annealing / basin-hopping / MCMC proposals; the certified-kernel reading sits in the Hamiltonian-Monte-Carlo (Duane et al. 1987; Neal 2011) and Metropolis-adjusted-Langevin (Roberts & Tweedie 1996) lineage (the squeeze a linear, gradient-free, mass-preconditioned replacement for the leapfrog trajectory; σ^* an FDT calibration, not a new algorithm); the coset-erosion result is Mermin–Wagner-flavoured; and the relativistic cap has a physics precedent in Lieb–Robinson light-cone bounds (Lieb & Robinson 1972). We cite these for the bound and lineage and claim only the *design consequences* — the $\det J = 1 + \text{energy-ledger} + \text{latch-transport certificate}$, and the governor-composed, coset-projected, FDT-calibrated, bounded-injection acceptance with an explicit certified-retry erosion budget — not novelty of nonlocality, stochastic escape, MCMC itself, or the causal bound.

A Flag-provenance tables

All results inherit the exact non-default configuration in effect. Repositories read-only for the analysis reports; branches for the code-producing reports named in Appendix A.4.

A.1 §3 certificate stack (paid-access-theory, paid-access-experiments)

flag	value
theory commit	9a13455 (analytic checks, numpy float64, <code>default_rng(0)</code>)
experiment commit	6f2384c (branch tip <code>agent/experiment-engineer/paid-access-experiments</code> , off <code>main@63fea62</code> ; not pushed)
JAX	0.9.0 (main venv reused, protocol §4)
kinetic modes	relativistic (reach, wormhole, throat); <code>newtonian_learned</code> (Newtonian control, injection cert)
mass band (prerequisite)	$[4.0, 0.25] \Rightarrow M_{\text{eff},0} = 4.0$ (heavy reach coord), $16\times$ contrast $\Rightarrow S^{(M)}$ directional
c / rest mass / m_0	1.0 / 1.0 / 1.0
dt / T (reach horizon)	$0.05 / 100 \Rightarrow L = T\epsilon c / \sqrt{M_0} = 2.5$, $v_{\text{max},0} = 0.5$
γ	reach rollout $\gamma = 0$ (sharp box); governor re-absorption not γ -swept
ζ grid	$[0, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0]$ (line-searched; success = any ζ lands)
basin geometry	double well along coord 0, wells $\{0, d\}$, barrier $\Delta V_b = 1.0$, $d \in \{0.8, 1.6, 2.4, 3.2, 4.0, 5.0\}$
init	$q_0 = 0$, $p_0 = (1.2, 0, \dots) + 0.02\mathcal{N}$ ($\text{KE}_0 < \Delta V_b \Rightarrow$ plain relax escape-blocked by design)
landing criterion	$\min_{\text{traj}} q_0 - d < 0.4$
seeds	reach $\{0, 1, 2, 3, 4\}$; latch/injection <code>default_rng(0)</code>
dims	2 (headline) and 4 (identical result)

A.2 §4.1 calibrated gate (v1-pivot; memory-agnostic transfer from minus-the-physics Part B)

The gate/calibration/LTT numbers are from `v1-pivot`. The **memory-agnostic** finding (CM-2 approved wording) — the identical stack on a modern Hopfield memory reproduces the calibration jump, raw $0.18 \rightarrow 0.88$ vs CLU $0.43 \rightarrow 0.87$ — is from `minus-the-physics` Part B (5 seeds), whose provenance is inherited here.

flag	value
commit	572c708 (branch <code>agent/experiment-engineer/v1-pivot</code> , off <code>main@dbeb2c2</code> ; not pushed)
Hopfield-transfer source	<code>minus-the-physics</code> Part B, 5 seeds (same MQAR vocab-256, $\text{kv} \leq 32$ probes; CM-2)
task	MQAR CLU-EBM, vocab 256, $\text{kv} \in \{16, 24, 32\}$, 128–144 trials·seed ⁻¹ ·level ⁻¹
seeds	5 (base 42); 90 models, 609s CPU
write	kinetic relativistic, potential mlp, hidden 128, epochs 400, lr $1e-3$, batch 16, k_steps 50, buffer 128, friction 0.3, temperature 0.3, input noise σ 0.05, PCD
gate	calib features <code>r_margin</code> , learned τ at write time, $p_{\text{exit}} = 0.5$; cost checkpoints 300/1200/2100/3000
LTT	fixed-sequence, exact binomial p -values, targets $\epsilon \in \{0.05, 0.1\}$
Hopfield baseline	Platt-calibrated logit margin, same probes
<code>langevin_noise</code>	legacy (default); lyapunov N/A (PCD write, no MSE/Lyapunov)

A.3 §4.2 routing (v1-router-baseline, v1-wormhole-routing)

flag	value
commit	52330f8 on 9339a13, off main@9a13455 (branch agent/experiment-engineer/v1-router-baseline; not pushed)
only non-default flag	experiment.v1.wormhole.n.seeds=5 (default 2)
lattice/task	$N \in \{4, 8\}$, embed_dim 12, embed_scale 2.0, vocab 128, kv_per_unit 3, query_cue_noise 0.05
write	kinetic relativistic, potential mlp (coercive), hidden 128, epochs 400, lr 1e-3, batch 16, k_steps 50, buffer 128, friction 0.3, temperature 0.3
retrieval/routing	dt 0.05, relax_steps/route.steps 250, governor_sensitivity 0.95, $\kappa_{\text{wh}} = \kappa_{\text{chain}} = 2.0$, gate.z.threshold 0.0, gate.z.width 0.7, gate.route.threshold 0.5
router MLP	hidden 32 (449 params), epochs 300, lr 3e-3, l2 1e-3
FLOPs model	flops_grad_factor 6.0, flops_verlet_grads 2.0; routed leg = 2 units (flat in N), chain = N units
workload mixes	[[.5, .5], [.8, .2], [.95, .05]]

A.4 §4.3 regime map (v1-hopfield-stress, regime-remap-2000ep, anchor-robustness item 2)

flag	value
commit	63fea62 (main, w6-integrated); analysis read-only, artifacts under .claude/
experiment	experiment.v1.gate regime map; dtype f32
train_epochs	{500, 1000, 2000, 4000} (the arm)
kinetic / potential	relativistic / mlp (coercive)
embed_dim / hidden / vocab	16 (CLU dim 32) / 128 / 256 (512 for kv128)
retrieval	relax_steps 300, governor_sensitivity 0.95, calib r_margin, $p_{\text{exit}} = 0.5$, cost ladder 300/1200/2100/3000
Hopfield	$\beta \in \{2, 5, 20\}$; iteration sweep {1, 2, 3, 5, 10} (Item 3)
langevin_noise	legacy (FDT-violating default; kept for baseline continuity)
seeds	{42-46} (Item 1) / {42, 43, 44} (Item 2) / {42, 43} (anchor item 2, 2 episodes/cell)

A.5 §3.2.1 latch payoff + Appendix B.2 BIBO battery (v1-certificate-payoff)

These runs extend the §3 battery with two new arms; the defaults preserve the §3.1/§3.2/C.1 numbers **bit-identically** (reach table, latch transit, and squeeze injection all reproduce paid-access-experiments to the digit).

flag	value
commit	27f232f (branch tip <code>agent/experiment-engineer/v1-certificate-payoff</code> ; payoff code <code>d9a9f38</code> , config <code>f2a85aa</code>); base local <code>main@37dc664</code> ; not pushed
JAX	0.9.0 (main venv reused, <code>--no-sync</code> ; protocol §4)
training	none — analytic potentials, oracle channel placement (learned entrance-steering out of scope)
kinetic mode	relativistic (reach, wormhole, throat, BIBO); newtonian_learned (Newtonian control, injection cert)
mass band (prerequisite)	$[4.0, 0.25] \Rightarrow M_{\text{eff},0} = 4.0$, $16\times$ contrast
$c / m_0 / dt$	$1.0 / 1.0 \Rightarrow v_{\text{max},0} = 0.5 / 0.05$
γ	reach: 0 (sharp box $L = 2.5$); BIBO : 0.02 (bounded arms must settle)
BIBO potential	$k = 1.0$, $\epsilon = 0.02 \Rightarrow x_b = 3.5355$, $V_b = 3.125$; transverse conf = 4.0
BIBO exits / horizon / escape	$b \in \{1.0, 2.0, 3.0, 3.6, 4.0, 5.0\}$; $T = 2000$ (r^* also at $2T = 4000$); escape radius 20.0; $p_0 = 0.3$; margin 10^{-3}
latch payoff	16 incoming states, ball radius $0.3 < \rho = 0.35$; $f = 3.0$, $p_{\text{latch}} = 0.5$; $\ \Delta\ = 0.5$
seeds	reach/BIBO $\{0, 1, 2, 3, 4\}$; latch cloud <code>default_rng(0)</code> ; robustness seed ₀ $\in \{0, 7\}$
dims	2 (headline) and 4 (identical structure)
tests	<code>pytest tests/test_paid_access.py tests/test_core.py</code> $\rightarrow$ 16 passed (7 paid-access incl. 2 new regression tests + 9 core)

Cross-section reproducibility note (C-7). The two configuration axes a reviewer must track are **damping** and **training epochs**. *Damping*: §4.1 uses γ /friction 0.3 at write, PCD; §3’s reach rollout uses $\gamma = 0$ for a sharp box, while Appendix B.2’s BIBO battery uses $\gamma = 0.02$ **because a bounded arm must be able to settle** (at $\gamma = 0$ a conservative orbit never converges, so “bounded” would be untestable); §4.3 uses `langevin_noise=legacy`. *Epochs (the axis that governs the §4.1 $\leftrightarrow$ §4.3 comparison)*: **§4.1 trains 400 ep; §4.3 sweeps $\{500, 1000, 2000, 4000\}$ ep**. These are different budgets, and the paper states so at the point of collision: §4.1’s escalatable-accuracy headroom is a 400-ep (not-yet-converged) property, and §4.3, App. C.4.a gives the converged (2000-ep) counterpart where the gate’s accuracy *matches* full-budget and the payoff is rationing only. The two are reconciled inline in §4.1 point 2, not asserted away here.

B The paid-access certificate table, and the two fine-print experiments

B.1 Table 1 — the receipt per mechanism

mechanism	energy injected (bound)	volume det J	governor absorption	re-	latch impact
wormhole (translation)	$\Delta V = V(b) - V(a)$, discrete, ledgered	1 exactly; $(1-\gamma)^d$ damped	$\approx (1/\gamma_c) \ln(1+\Delta V/E^*)$		transport , shift $p^\top X \Delta$
wormhole (throat in V_θ)	standing $\sim A$, no jump	$(1-\gamma)^d$ exact	continuous		adds $\mu^2 \propto A/\ell^2$
squeeze $S^{(M)}$	$\leq (e^{2 \zeta }-1)H$ (matched quad. H)	1 exactly	$\approx 2\zeta/\gamma_c$		preserved (off-coset)

BIBO caveat (with the table). Both compose with a coercive V_θ and $\gamma > 0$ (both $\det J \leq 1$). But Deep/Conv potentials are non-coercive out-of-unit; a wormhole exit into a non-coercive region can escape to infinity. **Design constraint:** exits inside a coercive sub-level *component* (or with $\alpha\|q\|^2$ confinement). This is now **exercised and measured** (B.2); the §3.2 reach testbed uses coercive wells, where the constraint is inactive.

B.2 The BIBO battery — a free ledger does not buy boundedness

Reporting grade: verification (designed analytic testbed; oracle placement). Source `v1-certificate-payoff` §7.4. We make the architectural non-coercivity of Deep/Conv potentials analytic and controllable with $V = \frac{1}{2}kq_0^2 - \epsilon q_0^4 + \frac{1}{2}\text{conf}\|q_1\|^2$ ($k = 1$, $\epsilon = 0.02$, $\text{conf} = 4.0$), coercive *only* inside the connected component $|q_0| < x_b = \sqrt{k/4\epsilon} = 3.536$; barrier $V_b = 3.125$. Three arms, six exits b ; $\gamma = 0.02$, $T = 2000$, relativistic, 5 seeds; $r^* = \max_t \|q_t\|$ measured at T and $2T$.

arm (r^* at $2T$)	$b=1.0$	2.0	3.0	3.6	4.0	5.0
wormhole + receipt (screened)	1.01	2.01	3.01	0.09	0.09	0.09
wormhole, receipt ignored (ablation)	1.01	2.01	3.01	102.13	103.43	104.83
state-replacing map (no receipt)	1.01	2.01	3.01	102.13	103.43	104.83
escape rate, certified	0	0	0	0	0	0
escape rate, blind / router	0	0	0	1.0	1.0	1.0
wormhole receipt	ADMIT	ADMIT	ADMIT	REJECT	REJECT	REJECT
<i>energy-only sub-level test</i>	admit	admit	admit	reject	admit	admit

(T -values for the escaping arms: 52.12/53.43/54.83.)

1. **The diagnostic is a growth rate, not a big number.** $r^*(2T)/r^*(T) = 1.000$ for every bounded arm/exit (saturated) vs 1.96/1.94/1.91 for the escaping ones — $r^* \propto T$, genuinely unbounded, terminal velocity capped at $c/\sqrt{M_0} = 0.5$ by the relativistic governor. **The causal cap bounds speed, not excursion.**
2. **The receipt predicts BIBO blow-up on 6/6 exits;** the ledger $\Delta H = V(b) - V(a)$ is exact to 0.0 at every exit.
3. **The killer row is $b = 5.0$.** $V(5.0) = 0.0$ *exactly*: the ledger says the jump is **free** — cheaper than the admissible $b = 3.0$ exit ($\Delta H = 2.88$) — and an energy-only sub-level test **admits** it. It escapes anyway; likewise $b = 4.0$ ($V = 2.88 < V_b$; note $V(3.0) = V(4.0) = 2.88$ exactly, a coincidence of $\frac{1}{2}x^2 - 0.02x^4$). $\det J = 1$ **plus a bounded — or even zero — ledger is NOT sufficient for BIBO; coercive-component membership is the operative clause.**
4. **Attribution, honestly.** `wormhole_blind` and `no_physics_router` **coincide exactly**: neither is screened. So what buys BIBO is **the receipt, not the jump mechanism**. The defensible claim: *the wormhole can form the receipt* (it has V_θ , an exact ledger, a well-defined Jacobian) *and a physics-free state-replacing map has none of the three and cannot screen its own exit even in principle* — bolting a coercivity check onto it means handing it the certificate machinery. **Payoff A (§3.2.1) carries no such caveat, which is why it, not this, is the headline panel.**
5. **The trade the certificate makes explicit.** On rejected exits the certified wormhole **does not reach the target** ($r^* = 0.09$; it stays home). The receipt does not make an unsafe exit safe — it *refuses* it.
6. **Scope.** The coercive screen uses the analytic x_b : an **oracle** too. Certifying exits on a trained, non-coercive V_θ is genuinely open. Also $b = 3.6$ sits 0.06 outside x_b , yet $r^*(2T) = 102$ vs 0.09 and growth 1.96 make the classification unambiguous — no knife-edge, unlike the §3.2 squeeze bracket.

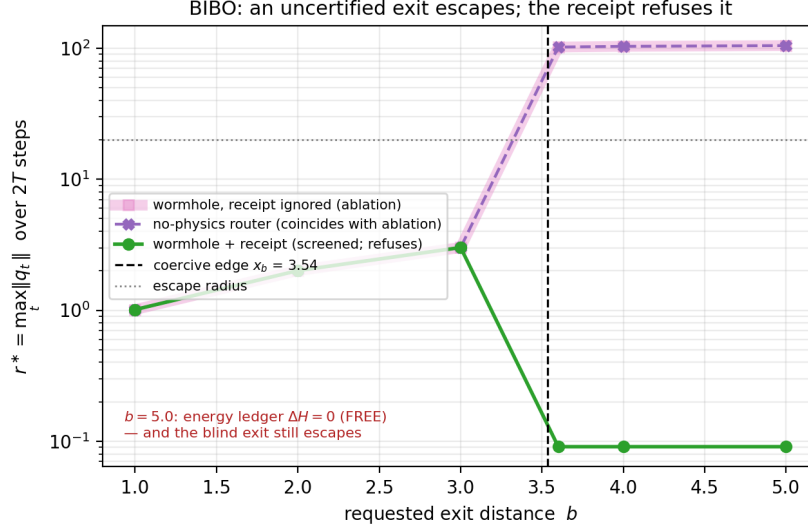


Figure 4: **BIBO battery (App. B.2, verification grade)**. An uncertified exit escapes ($r^* \propto T$); the receipt-screened exit stays bounded by *refusing* to jump. The router curve coincides *exactly* with the receipt-ignored wormhole ablation (drawn as a translucent band under a dashed overlay so neither is occluded) — the honest reading is that the *receipt*, not the jump, buys boundedness. Note $b = 5.0$: the energy ledger is free ($\Delta H = 0$) and the exit still escapes.

C Full honest tables

C.1 Reach battery, full grid (Table 2; paid-access-experiments §7.1, dim 2, 5 seeds, $L = 2.5$)

arm	$d=0.8$	1.6	2.4	3.2	4.0	5.0
plain_relax	0	0	0	0	0	0
squeeze $S^{(M)}$	1	1	1	1	$0^\dagger$	$0^\dagger$
wormhole	1	1	1	1	1	1
newtonian_squeeze (control)	1	1	1	1	1	1
state-replacing map (no_physics_router)	1	1	1	1	1	1
throat/dense- V	1	0.8	0	0	0	0

[†] The squeeze’s 0 at $d \in \{4.0, 5.0\}$ is “priced out of the swept $\zeta \leq 2.0$ grid,” not “cannot reach” (the pricing law, §3.2). The reachable radius is the bracket $[L, L + p_0 \sinh \zeta / M_0]$ with $L = 2.5$, $p_0 = 1.2$, $M_0 = 4.0$; the *same* verified bracket predicts $d = 4.0$ lands at $\zeta \geq 2.0105$ (energy $\approx e^{4.02} H$) and $d = 5.0$ at $\zeta \geq 2.6441$ (energy $\approx e^{5.29} H$) — reach beyond L is bought at exponentially rising energy, whereas the wormhole reaches all d at a fixed ledger. These two ζ values are analytic predictions from the verified bracket formula, not new measurements; the swept grid ended at $\zeta = 2.0$.

Note on the two “router” objects. The no_physics_router arm above is an *untrained analytic constant map* $(q, p) \mapsto (b, p)$ (training: none, A.5) — a **state-replacing** baseline, distinct in kind from §4.2’s router_mlp, a 449-param *learned decision head* that routes *through* the wormhole’s $\det J = 1$ edge. Only the former has $\det J = 0$; the latter inherits the wormhole receipt.

Certificates on every arm: wormhole $\det J = [1.0] \times 6$ (exact), ledger err $[0.0] \times 6$; **state-replacing map** $\det J = [0.0] \times 6$ (measured by forward-mode autodiff — the map $(q, p) \mapsto (b, p)$ has

Jacobian $\text{blockdiag}(0_d, I_d)$, so it is **volume-annihilating and non-invertible, not merely uncertified**); squeeze $\det S^{(M)} = 1.000 (\pm 4\text{e-}6)$, symplectic err $6\text{e-}8\text{--}1.3\text{e-}6$, H -ratio $\leq e^{2\zeta}$ (matched quadratic H); forbidden state-dependent gate $\det J = 2.05$ (design guard). Latch transit: zero-shift $\Delta Q = 0.0$; across-coset $\Delta Q = 0.2500 = p^\top X \Delta$; squeeze $\Delta Q = 1.2\text{e-}7$; random shift erases ($\det J = 1$ notwithstanding).

C.1.b — The latch-payoff cloud (§3.2.1), all arms. 16 incoming states, capture ball radius $0.3 < \rho = 0.35$, $\text{std}(Q_{\text{in}}) = 0.0803$; per-arm reconstruction error where defined.

arm	$\det J$	$\text{std}(Q_{\text{out}})$	$\text{std}(Q_{\text{out}})/\text{std}(Q_{\text{in}})$	ΔQ	std	reconstruction err
wormhole, coset-tangent	1.0	0.08029	1.0000002	$6.0\text{e-}8$		$2.2\text{e-}8$
wormhole, across-coset	1.0	0.08029	1.000000	0.0		(exact by construction)
random shift	1.0	0.27933	3.479	0.2465		— (no channel)
state-replacing map (no_physics_router)	0.0	0.0	0.0	0.0803		undefined (non-invertible)

Replication ($\dim \times \text{seed}_0$): $\text{std}(Q_{\text{out}})_{\text{wormhole}} = \text{std}(Q_{\text{in}})$ in every cell — 0.0803 (dim 2, s0), 0.0679, 0.0533, 0.0448 — and $\text{std}(Q_{\text{out}})_{\text{router}} = 0.00\text{e+}00$ in every cell.

C.2 Routing, full grid (v1-router-baseline, 5 seeds; acc local/distant, FLOP/query)

arm	$N=4$ acc (L/D)	$N=4$ FLOP	$N=8$ acc (L/D)	$N=8$ FLOP
local_only	0.500 (1.00/0.00)	5.88e7	0.500 (1.00/0.00)	5.88e7
gated (energy)	0.887 ± 0.139 (0.90/0.88)	1.18e8	0.715 ± 0.172 (0.82/0.61)	1.18e8
dense	0.500 (0.00/1.00)	1.18e8	0.448 ± 0.070 (0.00/0.90)	1.18e8
chain	0.652 ± 0.159 (0.90/0.41)	1.76e8	0.548 ± 0.138 (0.82/0.28)	2.94e8
calibrated (energy head)	0.860 ± 0.122 (0.72/1.00)	1.34e8	0.677 ± 0.174 (0.46/0.90)	1.49e8
router_mlp (no physics)	1.000 ± 0.000 (1.00/1.00)	8.81e7	0.948 ± 0.070 (1.00/0.90)	8.81e7

Per-mix: the router leads on accuracy AND FLOPs at every mix $\{50/50, 80/20, 95/5\}$ and both N . Salvaged mechanism: 1-hop routed leg flat 1.18e8; chain $1.76\text{e}8 \rightarrow 2.94\text{e}8$ with distant recall $0.41 \rightarrow 0.28$.

C.3 Regime map — Hopfield iteration parity (regime-remap-2000ep Item 3, 3 seeds)

cell	β	1 iter	2	3	5	10
N128/kv32	5	0.976	0.976	0.979	0.979	0.979
N256/kv64	5	0.967	0.969	0.969	0.969	0.969
N384/kv96	5	0.941	0.950	0.949	0.949	0.949

Hopfield reaches its ceiling in a single matvec at $\beta \geq 5$; $10\times$ more iterations change accuracy by ≤ 0.003 . Cost floor $O(\text{kv} \cdot d)$.

C.4 Regime map — settled full grid (regime-remap-2000ep, 198 jobs, 0 failures)

C.4.a — Capacity axis, paired 500 vs 2000 ep (corr = 0; $n = 8$ pooled seeds; Figure 2a–b). Δ = gate – Hopfield; savings = intra-CLU (full-budget CLU cost / gated CLU cost).

cell (N,kv)	ep	CLU fidelity	gate acc	full-budget acc	Hopfield acc	$\Delta(\text{gate-hop})$	intra-CLU savings
N128/kv32	500	0.76 ± 0.09	0.31 ± 0.04	0.30 ± 0.02	0.98 ± 0.01	-0.67	1.2×
N128/kv32	2000	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	+0.02	9.9×
N256/kv64	500	0.43 ± 0.05	0.06 ± 0.02	0.06 ± 0.02	0.97 ± 0.01	-0.91	1.0×
N256/kv64	2000	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.97 ± 0.01	+0.02	9.5×
N384/kv96	500	0.40 ± 0.03	0.02 ± 0.01	0.02 ± 0.01	0.95 ± 0.01	-0.93	1.0×
N384/kv96	2000	0.97 ± 0.01	0.91 ± 0.02	0.91 ± 0.02	0.95 ± 0.01	-0.04	6.2×

C.4.b — Epoch-scaling frontier $\{500, 1000, 2000, 4000\} \times \text{kv}\{32, 64, 96, 128\}$ ($n = 3$; `fig_frontier_clean.png`, plotted as **Figure 3** in §4.3). Hopfield band across these cells = 0.947–0.976 (epoch-independent, 1 matvec).

kv (N,vocab)	metric	500 ep	1000 ep	2000 ep	4000 ep	best ep (Δ vs Hop)
kv32 (N128,v256)	fidelity	0.74 ± 0.10	1.00 ± 0.01	1.00 ± 0.00	0.99 ± 0.01	
kv32	gate acc	0.30 ± 0.04	0.98 ± 0.02	1.00 ± 0.01	0.89 ± 0.05	2000 (+0.02)
kv64 (N256,v256)	fidelity	0.41 ± 0.02	0.82 ± 0.04	1.00 ± 0.00	1.00 ± 0.00	
kv64	gate acc	0.06 ± 0.02	0.42 ± 0.02	0.99 ± 0.01	0.97 ± 0.01	2000 (+0.02)
kv96 (N384,v256)	fidelity	0.39 ± 0.02	0.24 ± 0.06	0.97 ± 0.01	1.00 ± 0.00	
kv96	gate acc	0.02 ± 0.00	0.11 ± 0.01	0.92 ± 0.02	0.975 ± 0.00	4000 (+0.03)
kv128 (N384,v512)	fidelity	0.40 ± 0.02	0.09 ± 0.01	0.71 ± 0.06	1.00 ± 0.00	
kv128	gate acc	0.01 ± 0.00	0.04 ± 0.00	0.60 ± 0.06	0.95 ± 0.01	4000 (+0.004, tie)

The wall beyond kv64 is an **epoch-budget wall** (required epochs scale with kv: kv32 saturates by ≈ 1000 , kv64 by ≈ 2000 , kv96/kv128 by ≈ 4000), not a hard capacity limit. **Over-train (negative):** kv32 gate drops $1.00 \rightarrow 0.89$ from $2000 \rightarrow 4000$ ep. **Non-monotone fidelity dip (negative):** kv96 fidelity $0.39 \rightarrow \mathbf{0.24} \rightarrow 0.97$, kv128 $0.40 \rightarrow \mathbf{0.09} \rightarrow 0.71$ — a transient PCD/buffer instability at intermediate epochs; do not read the 1000-ep point as monotone progress.

C.4.c — Stress axes, 2000 ep ($n = 5$). $\Delta = \text{gate} - \text{Hopfield}$; **CLOSE** iff $\Delta \geq -0.01$.

Correlation- ρ axis:

cell	ρ	fid	gate acc	Hop acc	Δ	savings	closes?
N128/kv32	0.5	1.00	1.00 ± 0.00	0.98	+0.02	10.0×	✓
N256/kv64	0.5	1.00	0.99 ± 0.01	0.97	+0.02	9.3×	✓
N384/kv96	0.5	0.97	0.90 ± 0.01	0.95	-0.05	5.5×	no
N128/kv32	0.9	0.97	0.87 ± 0.06	0.72	+0.16	5.6×	✓
N256/kv64	0.9	0.89	0.67 ± 0.07	0.59	+0.08	2.6×	✓
N384/kv96	0.9	0.68	0.36 ± 0.05	0.52	-0.16	1.3×	no

$\rho = 0.5$ is identical to $\rho = 0$. At $\rho = 0.9$ Hopfield itself collapses (0.72/0.59/0.52) so the reversal *widens* at $\text{kv} \leq 64$ — Hopfield falling, not CLU rising (CLU gate also drops).

*Eval-noise- σ axis — **THE NOISE WALL** (the dominant negative):*

cell	σ	fid	gate acc	Hop acc	Δ	savings	closes?
N128/kv32	0.3	1.00	0.90 ± 0.03	0.94	-0.05	4.6×	no
N256/kv64	0.3	1.00	0.81 ± 0.03	0.89	-0.08	3.4×	no
N384/kv96	0.3	0.97	0.62 ± 0.04	0.85	-0.23	2.2×	no
N128/kv32	0.6	1.00	0.36 ± 0.06	0.71	-0.35	1.3×	no
N256/kv64	0.6	1.00	0.29 ± 0.04	0.57	-0.28	1.3×	no
N384/kv96	0.6	0.97	0.19 ± 0.01	0.52	-0.33	1.3×	no

Under noisy cues NO cell closes, at any capacity, even kv32 — CLU storage fidelity is still ≈ 1.0 , but the gated relaxation is far less robust to cue corruption than Hopfield’s single matvec. **Tally at 2000 ep: correlation axis 6/9 close (all $\text{kv} \leq 64$); eval-noise axis 0/6; overall 6/15.** Cost caveat repeated (attach to every accuracy sentence above): the savings are **intra-CLU** (gate

vs full-budget CLU, 3000 Verlet steps), not vs Hopfield, which remains the cheaper AND more cue-noise-robust retriever at matched accuracy (≥ 0.947 in ≈ 1 matvec).

D Negatives on the record

Tier-A V1 negatives (future-work anchors, not hidden failures): **N1** squeeze retries recover nothing at single-unit scale — *tested selection among reachable, not crossing to unreachable*; does not bear on §3 (§3.3). **N2** abstention-vs-Hopfield unwinnable; the 500-ep “Hopfield-dominant” map was an **under-training artifact** — at convergence the gate reverses Hopfield **only on clean/correlated cues at $kv \leq 64$** ($\Delta + 0.02$), the barrier beyond it is an **epoch-budget wall, not a capacity wall**, and the tally is **6/15** cells closing at 2000 ep (settled, full grid, 198 jobs). **N2b THE NOISE WALL: the accuracy reversal does not survive cue noise (dominant negative)**. Under eval-noise $\sigma \in \{0.3, 0.6\}$ **no cell closes at any capacity, even $kv32$** — gate 0.36 vs Hopfield 0.71 at $\sigma = 0.6/kv32$ (Δ from -0.05 to -0.35) — *despite* CLU storage fidelity ≈ 1.0 . The governed relaxation over-commits to the corrupted cue; the gate’s rationing is a **clean-retrieval** asset and **noise-robustness is Hopfield’s**. This is the axis most relevant to real retrieval and the sharpest negative in the study; a referee finding it absent would be right to fault the paper (App. C.4.c). Diagnosing/curing the noise-robustness gap (noise-aware τ , longer relax budget, denoising init) is future work. **N3** CLU energy $\approx$ readout margin ($\Delta \text{AUROC} \in [-0.004, +0.024]$) — *the forbidden energy-superiority claim, not made*. **N24** energy-gated wormhole loses to a physics-free router (the §4.2 boundary). **N30** $V(\text{data})$ -anchor does not transfer to memory fidelity. **N31** state-dependent jump-gate breaks volume by $\nabla g \cdot \Delta$ ($\det J = 2.05$; the frozen-gate design guard). **N23** smooth z -gate mis-routes ~ 10 – 12% . Scope caveat: N24/N27 rest on a linearly-separable-cue laptop testbed — honest for the tested regime, not proven universal.

Fine print of Payoff A — volume preservation alone is *not* the latch receipt (measured design guard). The random-shift arm is symplectic ($\det J = 1$) and *still* scrambles the Goldstone charge (ΔQ std 0.2465; out-spread $0.2793 = 3.48\times$ the incoming spread). Only the **matched channel** (Δ coset-tangent, $X\Delta \perp p$) makes ΔQ constant and preserves the spread. Any reading of §3.2.1 as “ $\det J = 1 \Rightarrow$ memory is safe” is wrong, and we flag it so the fine print cannot be inverted into the review.

Fine print of Payoff B — the receipt buys BIBO, *not* the jump; and the state-replacing map $\equiv$ a blind wormhole. The `wormhole.blind` and `no_physics_router` (state-replacing) arms coincide exactly on all six exits, so no boundedness advantage attaches to the wormhole’s jump mechanism per se — only to the screening receipt, which a physics-free state-replacing map cannot form (no V_θ , no ledger, no Jacobian). And a **free** ledger ($\Delta H = 0$ at $b = 5.0$) is admitted by an energy-only test and escapes anyway. This is why the latch (Payoff A), not BIBO (Payoff B), is the headline panel.

Payoff scope. Both payoffs are on the designed testbed: oracle channel placement, analytic potentials, no training. They demonstrate the certificate’s *downstream consequence*; they do **not** claim a real-data or learned-memory win. The coercive screen itself uses the analytic x_b and is likewise an oracle.

E Analytic verifications

`checks.py`, numpy float64, seed 0; double well $V = \beta(q^2 - a^2)^2$, $a = \beta = 1$, $\varepsilon = 0.05$. (A) reach cap $|\Delta q| = T\varepsilon c/\sqrt{M}$ to rel. err $2.0\text{e-}12$; (B) plain trapped $q = -1.068$, wormhole $q = +1.000$, ledger 0.0; (C) $\det J$ const = 1.000, state-dep matches $1 + \nabla g \cdot \Delta$ to $2.2\text{e-}16$; (D) squeeze hop threshold

$\zeta^* = 0.2356$ vs measured 0.27; (E) latch transit err $1.8e-15$, $X\Delta \perp p$ shift 0.0; (F) re-absorption 56 vs predicted 49.5 steps, injection $E_{\text{inj}}/E^* = 2.718 = e^{2.0.5}$.

F Test-time retries as a certified Markov kernel (derivations, checks, specified experiment)

Source: `thread9-mh-kernel` (4 numerical checks, float64, toy EBMs). **Grade: theory-complete on toy EBMs. No runs on trained CLU checkpoints are claimed anywhere in this paper.** This appendix carries the derivation behind the four design rules of §5.

F.1 Why the squeeze needs no Jacobian correction. $S_\zeta^{(M)}$ is exactly volume-preserving ($\det J = 1$) — the property HMC (Duane et al. 1987; Neal 2011) exploits to avoid a Jacobian correction in the acceptance ratio. A retry proposing a **sign-symmetrized** squeeze ($\pm\zeta$ with equal probability, restoring proposal symmetry) and accepting with $\min(1, e^{-\Delta H/T})$ is a **detailed-balance kernel for $e^{-H/T}$** — test-time compute as MCMC with a stationarity certificate. π -reversibility is proven and verified ($L_1 = 0.0095$ between forward and reversed two-step histograms). The certificate is **narrower than the retry-cascade pitch assumes**; three physics facts sharpen it.

F.2 (i) Squeeze-only is non-ergodic; MALA supplies mixing, FDT supplies its scale. $\{S_\zeta\}$ is a one-parameter subgroup whose orbit of any point is a **single hyperbola** in (q_i, p_i) : squeeze-only proposals are **reducible/non-ergodic** and cannot reach off-hyperbola states at all. Ergodicity requires HMC-style momentum refreshment; the squeeze is best read as a **mass-metric-preconditioned global (reach) move** layered on a MALA step (Roberts & Tweedie 1996). Its noise scale is not free: the discrete fluctuation–dissipation relation for the damped Verlet step fixes $\sigma_i^* = \sqrt{M_{\text{eff},i} T \gamma (2 - \gamma)}$, and this is **load-bearing rather than cosmetic** — the FDT-correct σ^* moves the sampler from shadow-biased to exact, L_1 error $0.0995 \rightarrow 0.0065$ (15 $\times$; float64). **Kinetic-mode scope (Newtonian only; a scope clause, not a retraction — V1’s units are Newtonian):** (a) once Metropolis-adjusted, σ^* is a proposal-tuning scale (mixing efficiency), not a correctness condition — any σ leaves π invariant, and the $0.0995 \rightarrow 0.0065$ gain is the finite-budget shadow bias removed; (b) the Gaussian FDT kick is a Gibbs momentum-refresh only in the Newtonian modes, where π_p is Gaussian — in a relativistic mode π_p is Maxwell–Jüttner, a fixed-covariance Gaussian refresh is not one of its refreshes, and stationarity is secured by the MH correction rather than by any σ^* . A mixture of two π -reversible kernels is π -reversible, so $\frac{1}{2}\text{MALA}(\sigma^*) + \frac{1}{2}$ squeeze-MH inherits the certificate.

F.3 (ii) The governor destroys the invariant: the real cascade is annealing, not sampling. State-dependent $\gamma(H) > 0$ is non- π -preserving dissipation, so the deployed squeeze-then-relax cascade is **Metropolis-within-annealing**, converging to a colder MAP-seeking measure: $T_{\text{eff}} : 1.0 \rightarrow 0.61$ as $\gamma : 0 \rightarrow 0.2$ (verified, float64). This may be *desirable for retrieval* — a MAP-seeking cascade is what an associative memory wants — but **it is not Gibbs sampling, and we never claim stationarity for the governed composite.** Hence design rule 1.

F.4 (iii) Decisive for a memory: even the certified retry erodes the latch. The Gibbs measure is **exactly flat along the Goldstone coset** (no restoring force along a broken continuous symmetry), so any coset-tangent proposal has $\Delta H = 0$ and is accepted w.p. 1. Accepted coset-tangent moves compose into an unbiased random walk that **diffuses the stored register** at

$D = \frac{1}{2}s^2$ per accepted move, with $N_{\text{erode}} \approx (\Delta_{\text{read}}/s)^2$, where s is the proposal scale and Δ_{read} the read-out margin along the coset [proven; verified $D = 1.29 \times 10^{-3}$ vs $\frac{1}{2}s^2 = 1.25 \times 10^{-3}$]. This holds **even for a charge-preserving isotropic squeeze**, which conserves $Q = p^\top X q$ while eroding coset *position* — charge and position are different observables and only the former is protected. Projecting proposals off the coset tangent quenches it ($D = 0$, verified), so only explicit writes move the register.

F.5 The deflation we state rather than bury. Acceptance is an *intra-model* quantity: cross-model incomparability of H does not break validity, but the operating temperature T that makes the kernel *useful* is model-specific and FDT-miscalibrated in general. A per-model T therefore **partially re-imports the learned gate** of §4.1 — the parsimony argument is weak. **The value of the MH framing is the stationarity certificate and the explicit erosion accounting, not parameter parsimony, and not, on current evidence, performance.**

F.6 The specified discriminating experiment (future work). Three arms (certified mixture kernel at $\gamma = 0$ with coset projection; the learned- τ escalation gate of §4.1; naive squeeze retries) on a trained CLU associative memory, matched on FLOPs, with **the latch-erosion decay curve as the money plot** — predicting that naive certified retries erode the register at $D = \frac{1}{2}s^2$ while the coset-projected kernel holds it flat. Risks: (a) the erosion rate is derived on toy EBMs and a trained- V_θ coset is only approximately flat; (b) the FDT σ^* assumes the damped-Verlet noise model, which `langevin.noise=legacy` runs violate; (c) matched-FLOPs accounting must charge the MALA gradient. **Partially answered, and on a different substrate:** the *mechanism* half — directed re-launch versus equal-energy random kick, ensemble-of- k restarts and ungated retry-all — is measured on a **designed** store (§5, design rule 2; MNIST). **What remains unrun is this specification itself: the certified mixture kernel at $\gamma = 0$ with coset projection, on a trained CLU checkpoint, with the latch-erosion decay curve. No trained-CLU-checkpoint result is claimed.**

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